

The VB5 interface

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The VB5 interface is an ATM-based broadband interface at the 'V' reference point, the interface between a user access and a service node (SN), also known as the SNI (service node interface) within the ITU-T. The B-ISDN architectural model allows for a number of different user accesses, such as straightforward digital sections and simple multiplexers, but it is anticipated that the most frequent implementation, particularly for mass deployment, will be via a broadband access network (AN) with VB5 interfaces.

The VB5 interface is specified in a similar way to its narrowband cousin V5, in two variants — VB5.1, which supports allocation of resources via the management system, and VB5.2, which adds a mechanism for allocating AN resources under the control of the SN, enabling call-by-call concentration in the AN. This paper describes the background to the development of VB5 and explains the two variants VB5.1 and VB5.2.

1. The background to VB5

The introduction of fibre transmission and multiplexing technologies in the local loop necessitated the development of digital access interfaces to the local exchange for telephony. These interfaces were initially proprietary but were eventually standardised in the early 1990s for telephony, integrated services digital network (ISDN) and leased lines. The European Telecommunications Standards Institute (ETSI) were first with their V5.1 [1] and V5.2 [2] interface specifications. The ITU-T Recommendations G.964 [3] and G.965 [4] followed, based on the ETSI specifications.

The current VB5 specifications [5, 6] draw on these earlier SNI specifications. Many of the higher level requirements are the same for broadband and narrowband networks and, consequently, the nature of their interfaces as well.

It is unfortunate that the term access network (AN) is often used to describe two different things. In general, the term access network covers the 'local' network between the customer and the local exchange. The other definition relates to a specific integrated and managed network, between a service node (SN) and the customer's premises. This type of AN is defined in ITU-T Recommendation G.902 [7] as: 'an implementation comprising those entities which provide the required transport bearer capabilities for the provision of telecommunications services between a service node interface (SNI) and each of the associated user/network interfaces (UNI)'. This is the definition of an AN used in the VB5 standards, and throughout this paper.

The general access network architecture and boundaries defined in ITU-T Recommendation G.902 are shown in Fig 1.

In addition to the SNI, each equipment has a Q3 management interface. It is a principle in G.902, and carried into VB5 specifications, that time-critical management functions (e.g. user port blocking) are performed by the AN system management function (SMF) and the SN SMF, communicating over the SNI. In the VB5 specifications a real-time management co-ordination (RTMC) protocol is defined for this purpose. All other management communication is via the relevant AN or SN Q3 interface.

The SNI shown in Fig 1 corresponds to the 'V' reference point in the ISDN protocol reference model. The parts of the broadband ISDN protocol reference model significant to VB5 are shown in Fig 2.

Where the access network is of the integrated type described in G.902, controlled by a common management system (as opposed to disparate isolated transmission systems, for example), the VB reference point is then referred to as VB5, as detailed in ITU-T Recommendation I.414 [8].

Two factors triggered the start of work on a VB5 interface, one being the thought that a broadband equivalent to the narrowband V5 interface would be necessary, and the other the liberalisation of telecommunications in Europe in 1998.

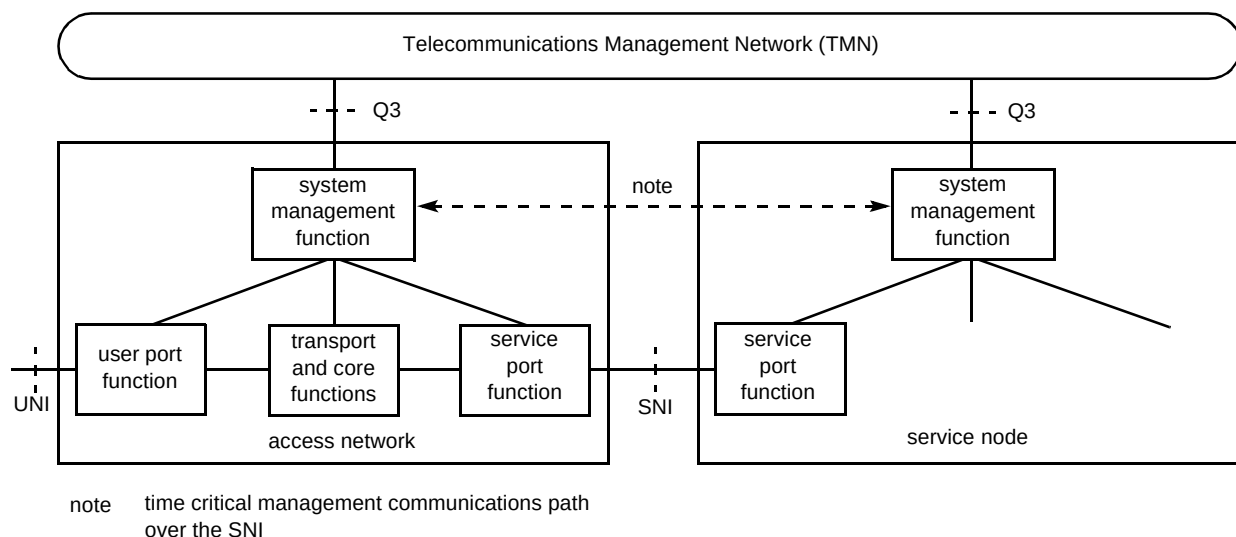


Fig 1 G.902 general access network architecture and boundaries.

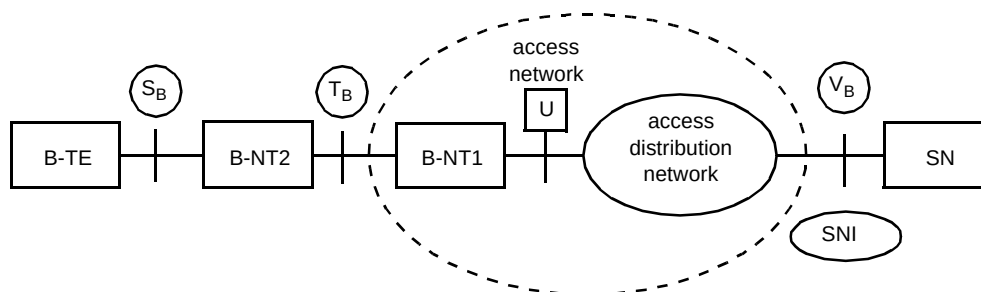


Fig 2 B-ISDN access protocol reference model.

The work in ETSI on specifying the VB5 interface began with a technical report (ETR) in 1994 [9]. At the time that this work started it was thought that probably there were already appropriate asynchronous transfer mode (ATM) specifications available, or in the process of being produced. A VB5 specification then need only consist of a set of these specifications, with particular options mandated. However, once the ATM specification issues were studied, this proved not to be the case.

One significant conclusion of the report was that two VB5 specifications should be produced:

- VB5.1, a relatively simple virtual path (VP) cross-connect-based version,
- VB5.2, a more sophisticated interface enabling connection-by-connection concentration in the AN under the control of the SN.

Each of these specifications would have a management specification associated with it to extend the AN and SN management models to include the VB5 specific aspects.

Many of the requirements identified in the report have been included in the current VB5.1 and VB5.2 specification, but, as there are some significant changes, the VB5 ETR is really now only of historical interest.

At the outset the principle requirements of a VB5 interface were that it should:

- be ATM based,
- be largely independent of the physical layer,
- be independent of user signalling protocols,
- provide the means for concentration in the AN,
- be scalable,
- support B-ISDN and non-B-ISDN access,
- provide broadband/narrowband integration,
- enable efficient business operations and maintenance processes,
- support shared UNIs (i.e. one UNI may be connected to a number of different SNs via separate SNIs),
- be specified with open network provision (ONP) in mind.

2. The VB5 interface

The VB5 interface is specified principally at the ATM layer. The only physical layer constraint at the VB5 reference point is a requirement to provide an embedded transmission path identification method (e.g. the path trace mechanism in synchronous digital hierarchy (SDH) systems). Any path protection must also be done, at the moment, at the physical layer, though VB5 specifications will probably be updated to include VP-level protection once a specification is stable. Figure 3 shows the functions at the VB5.1 and VB5.2 reference points. The broadband bearer channel connection (B-BCC) protocol is only used with VB5.2.

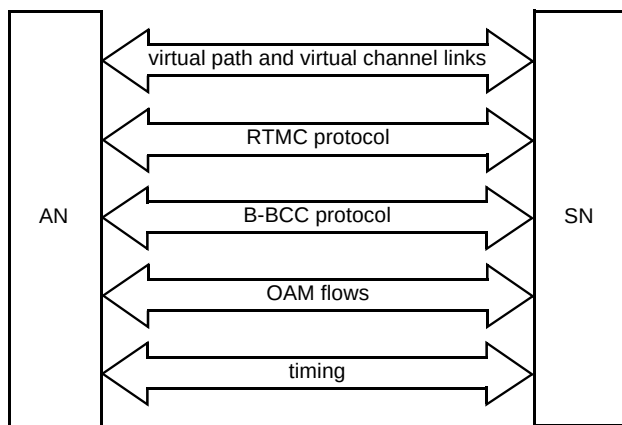


Fig 3 Functions at the VB5 interface.

At the VP level the normal ATM operations, administration and maintenance (OAM) procedures, specified in ITU-T Recommendation I.610 [10], are required, e.g. generation of VP-AIS/VP-RDI (alarm indication signal/remote defect indication). The cell header format and encoding, and the pre-assigned headers for use by the ATM layer used at the VB5 reference point, use the network-to-network interface (NNI) option of ITU-T Recommendation I.361, i.e. 12 bits of VP address space is available. It is assumed that if the generic flow control (GFC) field at the UNI is used, it will be handled within the AN itself, and not transported over the SNI.

2.1 General application of the VB5 interface

The general application of the VB5 interface is shown in Fig 4, which illustrates that access networks and service nodes can have multiple VB5 interfaces. These can be interconnected via a transmission network whose management is entirely separate from either Q3(AN) or Q3(SN). The transmission network can be anything from station cabling to complex SDH or ATM networks. Since the VB5 interface is specified at the VP layer it is possible to have multiple VB5 interfaces on the same physical link. The converse is also possible, a single VB5 interface may comprise multiple physical links.

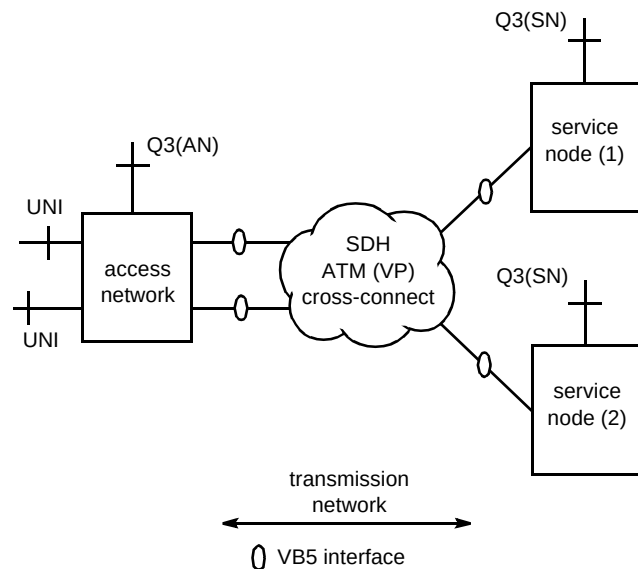


Fig 4 VB5 general application.

The VB5 interface is specified to allow single physical UNIs to have access to multiple service nodes. In fact, the support of shared UNIs is a requirement from G.902 and meets a European Commission ONP requirement that a user, served by an access network, should have access to multiple service nodes so that the supply of service need not be a monopoly. UNIs can also consist of multiple physical links.

Since the VB5 interface is only specified at the VP layer, some additional terms are required to identify specific groupings of VPs which comprise either a logical or physical interface (see Table 1). Identifiers related to these terms are used with certain VB5-specific protocol messages as well as the management model.

Table 1 VB5-specific logical and physical ports.

Logical/physical port		Definition
LUP	Logical user port	The set of VPs at the UNI or at a virtual user port (VUP) associated with one single VB5 reference point.
PUP	Physical user port	The physical layer functions related to a single transmission convergence function at the UNI.
LSP	Logical service port	The set of virtual paths (VPs) at one VB5 reference point (i.e. associated with one and only one service node) carried on one or several transmission convergence functions.
PSP	Physical service port	The physical layer functions related to a single transmission convergence function at the VB5 interface.

The relationship between the additional VB5 specific logical and physical ports is shown in the VB5 functional model in Fig 5.

In order to identify VPs in multi-link physical interfaces at both UNI and SNI, an extended VP addressing scheme is used which introduces a virtual path connection identifier (VPCI). This has a range of 16 bits. When used at the NNI it allows up to 16 physical interfaces per SNI (4 bits/interface and 12 bits/VPC). When used at the UNI it potentially allows 256 physical interfaces (8 bits/interface and 8 bits/VPC).

The number of VPCIs at both the SNI and UNI can be extended, if necessary, by reducing the number of bits allocated to the VP range and reusing those released to increase those available to identify the physical port. This requires co-ordinated provisioning in both AN and SN to ensure both entities are using the same addressing scheme.

The use of VPCIs is derived from the Digital Subscriber Signalling System No 2 (DSS2) [11]. At the UNI they are the same as those used in DSS2 signalling messages.

The relationship between the various connection element identifiers used in VB5, including the VPCI, is illustrated in Fig 6.

2.2 Narrowband support

VB5 interfaces provide for the integration of broadband and narrowband SNIs. This is achieved using a conventional ATM circuit-emulation technique to encapsulate the narrowband V5 or V3 framed 2048-kbit/s serial data.

2.3 Non-B-ISDN accesses

In addition to the standardised broadband ATM interfaces and legacy narrowband interfaces, there are a number of other user interfaces that different network operators may require. These are supported by VB5 using a virtual user port (VUP) concept. Essentially, at some logical

point in the AN, a UNI is assumed but not realised physically. This part of the AN is visible to the VB5 specific management model and handled in a generic way. Additional adaptation is performed in the AN beyond the VUP, but this is not considered part of the VB5 management model (see Fig 7).

This technique allows the AN to support any user interface that can be adapted to ATM, over a VB5 reference point.

2.4 Connection types

A number of different VP and virtual circuit (VC) connection types are defined in VB5 specifications to identify the types of connections that can be supported via a VB5 reference point to an AN. These are classified by the termination point, point or multipoint type and whether they are semi-permanent or switched.

VB5.1 can only support semi-permanent (VP and VC) connections directly. VB5.1 supports switched VCs as far as users are concerned, which are carried in semi-permanent VPs over the VB5 reference point. In this scenario a VP at the UNI maps directly to a VP at the SNI.

The AN can also provide unidirectional (SN-to-AN) point-to-multipoint connections via the VB5 reference point, i.e. the AN providing the cell-replication function. The inverse multipoint-to-point connections or bi-directional multipoint connections cannot be supported over the VB5 interface directly. These connection types can be provided to end users via point-to-point links over the VB5 reference point. It is the SN which must provide any combining function.

VB5.2 with its B-BCC protocol can support switched VCs directly (either bi-directional point-to-point or unidirectional point-to-multipoint), in addition to all of the connection types supported by the VB5.1 interface.

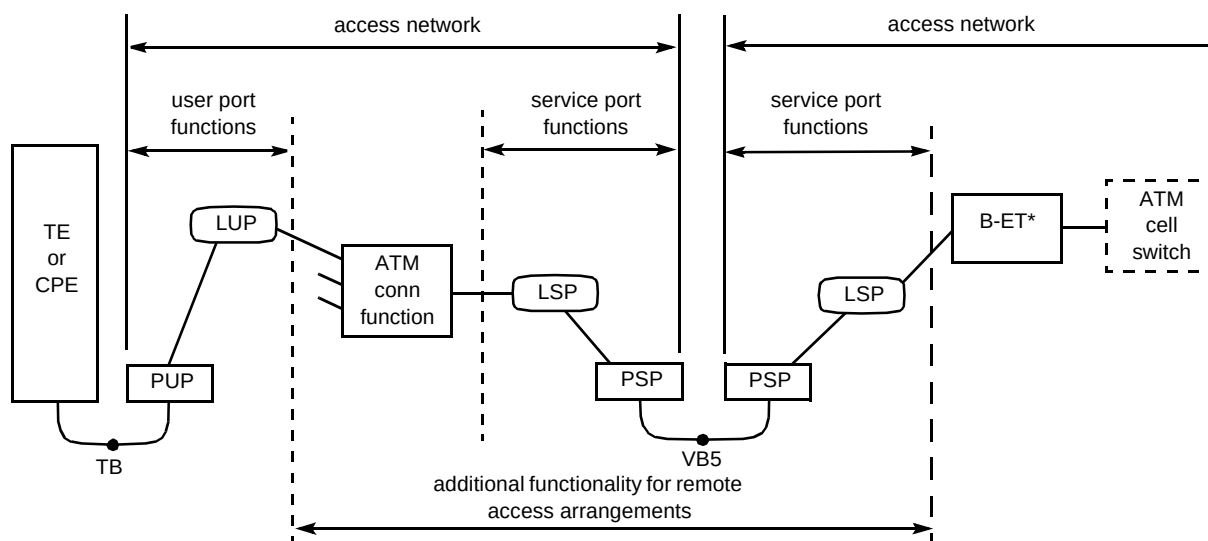


Fig 5 VB5 functional model.

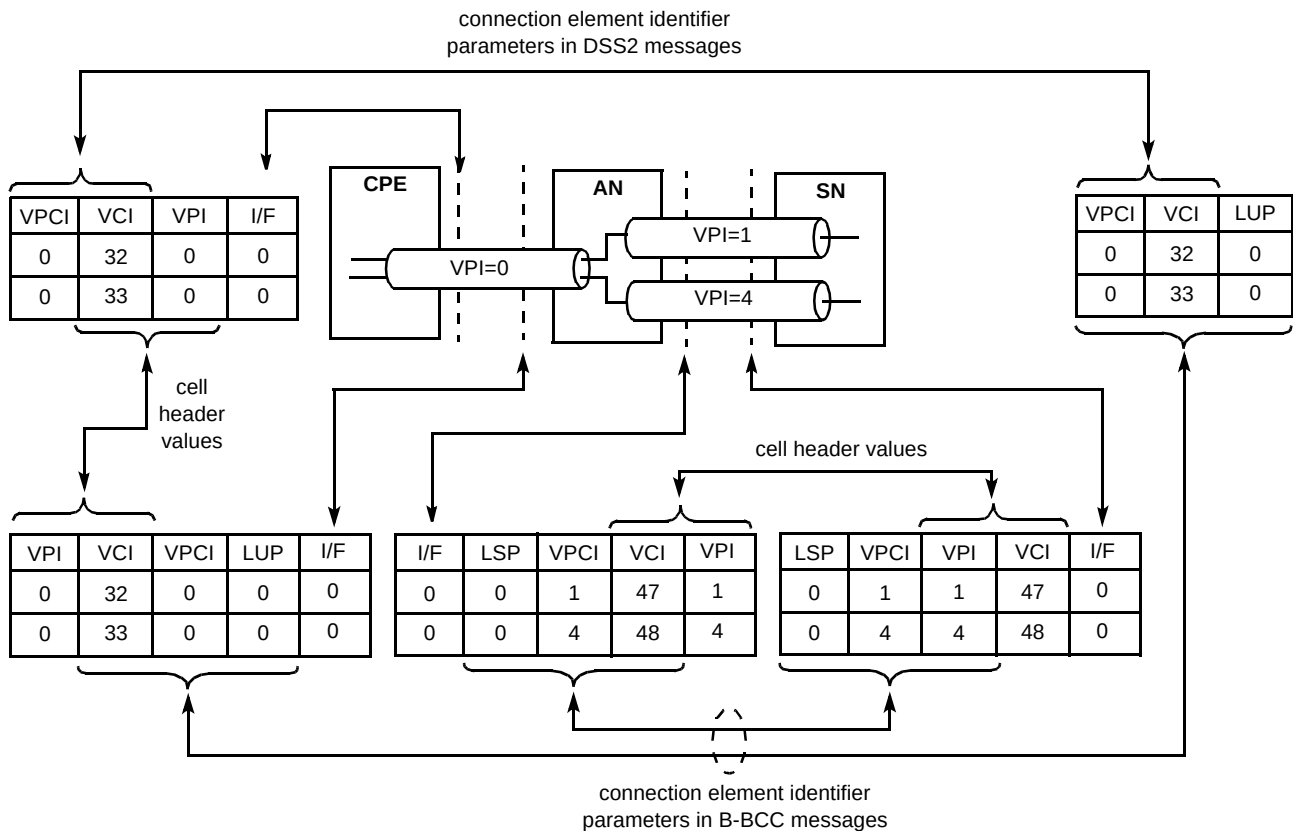


Fig 6 VB5 connection element identification.

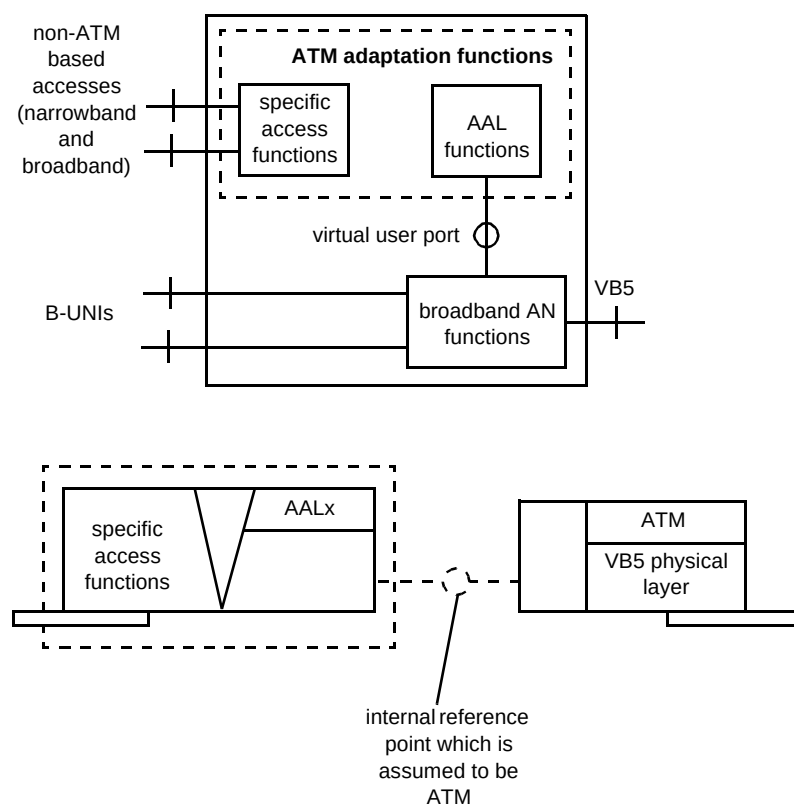


Fig 7 Virtual user port.

3. The real-time management co-ordination (RTMC) protocol

The main purpose of the RTMC protocol is to communicate real-time management information from the AN to the SN about the availability of AN resources which have an impact on the availability of service to the end users. In practice the change in status of only two types of AN resource is communicated — VPCs and the LSP. Other AN resources, such as PUPs, can be decomposed into sets of VPCs. Mechanisms are provided for the AN to indicate locked VPCs due to faults or administrative procedures. In the case of administratively locked VPCs there are two modes available — one stopping all connections, the other allowing only test connections.

A further shut-down procedure allows the VPCs to be locked gracefully, by waiting until all switched connections on a given set of VPCs are released, before locking the resource.

In addition to VPC status procedures, there are some further housekeeping functions provided:

- VPCI consistency check — ensures alignment of AN and SN configurations,
- LSP ID check — confirms matching interface identities either side of the SNI,
- RTMC reset — re-initialises the RTMC protocol entities.

3.1 ATM adaptation layer for the RTMC protocol

The RTMC protocol is carried in a VC in a reserved VP connection (VPC) between the AN and SN. The VP identifier (VPI) and VC identifier (VCI) values of this channel are not fixed and must be provisioned. The VPI range is unrestricted but the VCI cannot use any of the reserved values between 0 and 31. It was originally intended to have reserved values for the VB5 protocols. Some manufacturers were concerned about accessibility of VCIs less than 32 using current hardware. As it was also proving difficult to convince the relevant ETSI technical committee of the need to modify an existing standard to provide these values from a reserved set, they were made provisionable in the VB5.1 standard.

ATM adaptation is performed using the signalling AAL (SAAL) which comprises AAL5, SSCOP [12] and SSCF [13]. This is a specific ATM adaptation layer optimised for signalling applications also used in DSS2. Not all of the features available from these standards are required for the relatively simple RTMC protocol.

3.2 Port block/unblock and shut down procedures

The block/unblock procedure is used by the AN to inform the SN about the service relevance of the

administrative actions taken in the AN and to indicate service relevant fault conditions occurring in the AN.

This procedure is used in addition to the use of F4 OAM flows (e.g. VP-AIS), as described in ITU-T Recommendation I.610. There are a number of reasons why it was not possible to use only the facilities provided in I.610, the principal one being some ambiguity as to the application of VP-AIS when resources are locked. The responsible standards group was unable to address these issues in the necessary time-scales. There are other requirements related to resource locking which cannot be met by I.610 procedures, but which the RTMC provides:

- graceful 'shutting down' of resources,
- VP link status for user VPs terminated in the AN,
- status of unidirectional and point-to-multipoint VPs.

The type of lock is indicated by reason codes in a block message. The following reason codes are used:

- fully administratively locked (admFull) — the resource is not available due to administrative actions in the AN,
- partially administratively locked (admPart) — the resource is not available for normal traffic due to administrative reasons in the AN, but available for test calls,
- fault/error condition (Err) — the resource is not available due to a fault condition in the AN.

3.3 LSP ID Check

The purpose of the LSP ID check is to confirm that the AN and SN RTMC protocol entities are communicating with the correct peer entity. The actual LSP identity is a provisioned value, selected so that it is unique in any potential AN/SN combination.

The procedure is invoked automatically when the interface starts up and can be invoked from either Q3(AN) or Q3(SN) when the interface is operational.

3.4 VPCI consistency check

The VPCI consistency check is a procedure inherited from the core network broadband ISDN signalling user part (B-ISUP) of Signalling System No 7 [14]. The purpose of the VPCI consistency check is to ensure that VPCIs provisioned in the AN and SN are consistent. Any differences between the VB5 procedure and B-ISUP are due partly to the asymmetrical nature of the SNI and the fact that in VB5 the maintenance data is transported by a dedicated management protocol, the RTMC.

The procedure makes use of OAM loop-back cells to check the physical path and the RTMC protocol to collect information in the SN about the reception of these cells in the AN. The message sequence chart in Fig 8 illustrates the process for a single VPCI check.

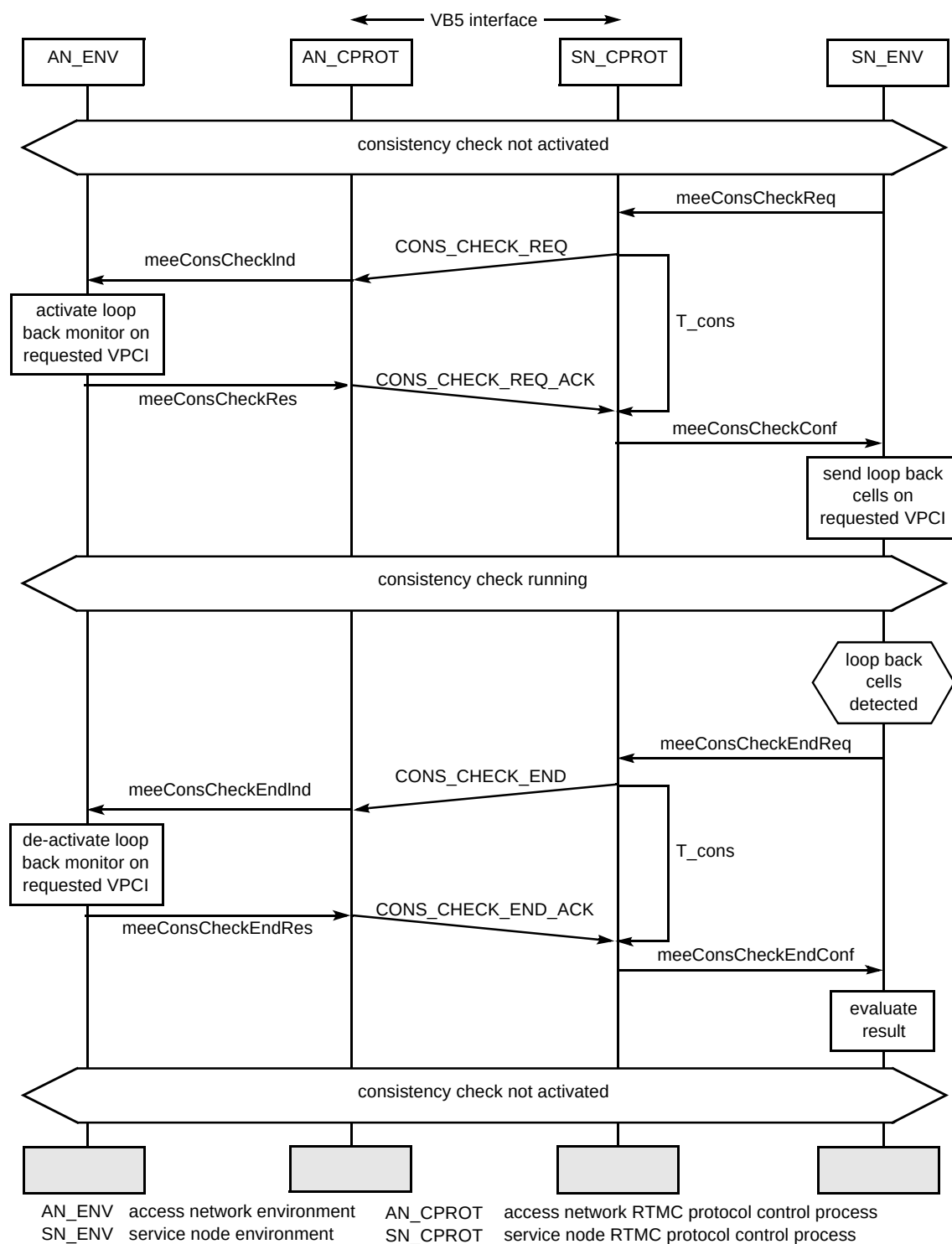


Fig 8 VPCI consistency check — message sequence chart.

The process can only be started by the SN. The SN begins by sending a message to the AN requesting that the OAM monitor be attached to the VPC. Once the AN has acknowledged the request, the SN sends some loop-back cells to the AN. When these are received back in the SN, it requests the AN for the result. The AN removes the loop-back monitor and replies. If the AN has seen some loop-

back cells it indicates success in the reply. The SN then moves to the next VPC.

The application of the VPCI consistency check is limited to VPCs which terminate in the access network. The current specification does not make use of OAM segments so it is possible only to use VPC end-points. If end-points

exist outside the AN, for the test to be performed, the end-point would have to be available (i.e. in the case of CPE, powered-up) and have a method of communicating the reception of loop-back cells on specific VPs, which is not practical with existing standards.

The alternative, using VP OAM segments, was considered. The principal limitation is that, since segments cannot overlap, once a segment is defined across the SNI for the VPCI consistency check, other segments crossing the SNI cannot then be used, i.e. from points inside the AN to the SN or beyond.

3.5 Reset

The only resources which can be reset are VPCs. These can be reset individually or all the VPCs in the LSP can be reset with one message. The purpose of the reset is to re-synchronise the VB5 specific finite state machines relating to the RTMC protocol on both sides of the interface. In order to speed resumption of service, the initial state of the VPCs is set to unblocked. Any VPCs which were actually blocked in the AN are subsequently re-blocked in the SN by messages from the AN.

4. The broadband bearer connection control protocol

The principal difference between VB5.1 and VB5.2 is the provision of a B-BCC protocol in addition to the RTMC protocol used with the VB5.1 interface. Any other differences between VB5.1 and VB5.2, such as the start-up procedure, occur only as a consequence of this additional protocol.

At the time of writing, the VB5.2 interface is still in the process of being defined in both ETSI SPS3 and the ITU-T Study Group 13. It is therefore quite possible that some detailed aspects of the VB5.2 interface described here will have changed by the time either the ETSI standard or ITU-T Recommendation is published.

The B-BCC allows the SN to control cross-connections in the AN and hence enables call-by-call concentration in the AN. Such concentration allows a larger number of UNIs to be supported by an AN for a given amount of SNI bandwidth. The general principle is in line with ITU-T Recommendation G.902. A similar technique is used in the narrowband V5.2 interface [2, 4].

There are a number of advantages that concentration in the AN provides using this technique:

- the AN only has to handle one relatively simple signalling protocol, since only bearer channel control is required,
- a minimal number of signalling interfaces are required in the AN, compared to the large number required if user signalling was terminated in the AN,
- no 'user profile' database is required in the AN,

- no routing information is required in the AN,
- the AN remains independent of any specific user/network signalling protocols,
- utilisation of the SN physical interfaces is optimised, since traffic is concentrated by the AN.

The B-BCC protocol uses the same type of SAAL as the RTMC protocol but does not share the same channel. A common VB5 protocol discriminator is used with both RTMC and B-BCC protocols. The message codepoints of each protocol are therefore defined so that they do not overlap. Figure 9 shows the general functional architecture of the AN from a protocol perspective.

4.1 B-BCC messages

The VB5.2 B-BCC protocol message set is summarised in Table 2. These messages are unique to VB5 but make use of information elements from DSS2, whenever practical, to reduce any interworking overhead. The initial concept of the B-BCC anticipated principally only the allocation/de-allocation type messages. However, to interwork with DSS2 and B-ISUP it is necessary to be able to modify connection parameters during call set-up and data transfer phases, hence the update and modify messages. Other messages will be defined, to handle interface start-up and resets for example. At the time of writing, the details of these were still the subject of discussions within ETSI SPS3 and the ITU-T Study Group 13.

Table 2 B-BCC protocol messages.

Message	Direction	Function
Allocation	SN → AN	Allocate the AN part of a connection — includes ATM transfer capabilities and QoS information
Allocation complete	SN ← AN	Indicates successful AN connection establishment
Allocation reject	SN ← AN	Indicates unsuccessful AN connection establishment
Audit	SN → AN	Check an AN connection
Audit complete	SN ← AN	Return data on an AN connection
De-allocation	SN → AN	De-allocate (release) an AN connection
De-allocation complete	SN ← AN	Result of de-allocation
Modify	SN → AN	Modify AN connection parameters
Modify complete	SN ← AN	Acknowledge successful modify request
Modify reject	SN ← AN	Indicate unsuccessful modify request
Update	SN → AN	Update AN connection parameters
Update complete	SN ← AN	Acknowledge successful update request
Update reject	SN ← AN	Indicate unsuccessful update request

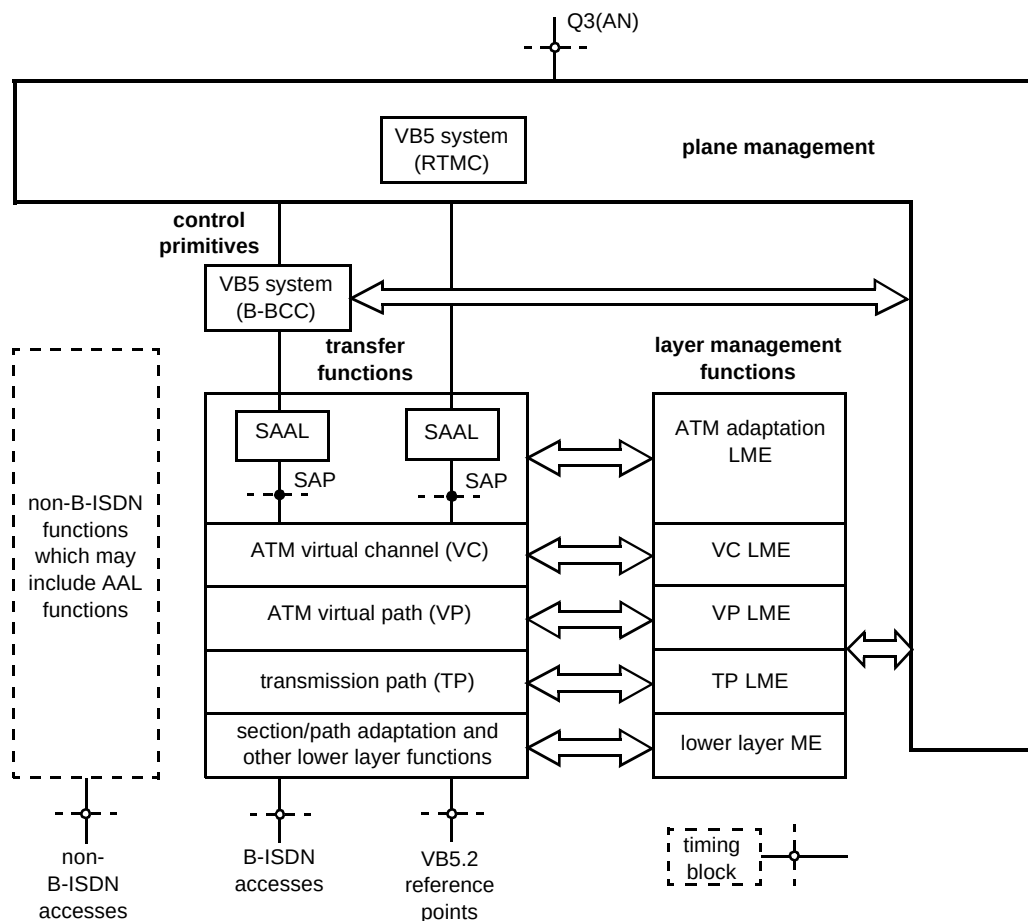


Fig 9 AN general functional architecture.

The B-BCC protocol is highly asymmetrical in that virtually all control rests with the SN. None of the connection-related transactions can be initiated by the AN.

4.2 Connection and transaction identifiers

Messages are identified by unique transaction identifiers. These have relevance only for a single request/acknowledge transaction. This is the same as the mechanism used with the RTMC protocol. To avoid having to identify both ends of a connection with the AN end-point resource identifiers (used originally to allocate the connection), a connection identifier is introduced. This is created by the SN, has the lifetime of a connection, and is first used in an allocation message. It is subsequently released back into the pool of available references with the de-allocation message. All transactions after initial allocation can use this method of referring to existing connections.

4.3 Connection establishment in the AN

The SN begins establishing a connection in the AN using the allocation message. Table 3 lists all of the possible information elements which can be used with an allocation message. The majority of these information elements are inherited from DSS2.

Table 3 Allocation message information elements.

Information element	Type	Comments
Connection reference number	M	Used over the lifetime of a connection.
ATM traffic descriptor	M	
Broadband bearer capability	M	
OAM traffic descriptor	O	
Quality of service descriptor	O	
User port connection identifier	M	VB5 specific addressing
Service port connection identifier	M	VB5 specific addressing
ABR set-up parameters	M	
End-to-end transit delay	O	For further study
Cell delay variation tolerance	O	For further study
Alternative ATM traffic descriptor	O	
Minimum acceptable ATM traffic descriptor	O	

Note: Information elements shown as type 'M' are mandatory, whereas type 'O' are optional.

In general, it is not possible to determine the final connection parameters immediately when the allocation message is sent, since other nodes in the connection may change the parameters as part of the negotiation during call set-up. An update message is therefore provided which allows the SN to revise the AN connection parameters, once

it has received the updated parameters negotiated with other nodes.

It is possible, with local calls, for example, where no other nodes are involved, that an update message would not be required. A message sequence chart in Fig 10 shows the use of the B-BCC messages in connection establishment, as part of the description of interworking with DSS2 and B-ISUP.

4.4 Connection admission control

Connection admission control (CAC) for the AN part of the connection is performed principally by the AN. The SN has some information, such as the availability of certain AN resources via the RTMC protocol, and so could decide not to invoke the B-BCC in some circumstances. The normal case though would be for the SN to request a connection and the AN to acknowledge or reject the request, having performed its own CAC.

A related issue is the allocation of VPI and VCI values. If the AN is performing its own resource management and

connection admission control, there is some benefit in allowing the AN to allocate VPI and VCI values. To achieve a similar level of optimisation, were the SN to allocate the resources, would require the SN to have detailed AN-specific data. This in turn would have to be defined in a generic way, to maintain an open interface. By keeping the data in the AN, manufacturers can optimise AN equipment without compromising the interoperability of the interface. However, a consequence is that the SN must be able to indicate more than just a single VPI/VCI value for connections in those situations where the AN can be offered a choice.

4.5 Interworking with DSS2 and B-ISUP

VB5 is not intended only to work with DSS2 and B-ISUP; however, it is anticipated that the most significant application of VB5.2 will be in networks which employ these user and network signalling protocols. It therefore makes sense to design the B-BCC such that interworking with DSS2 and B-ISUP is not unnecessarily complicated. The general design of the B-BCC is biased towards DSS2 rather than B-ISUP since architecturally the VB5 interface

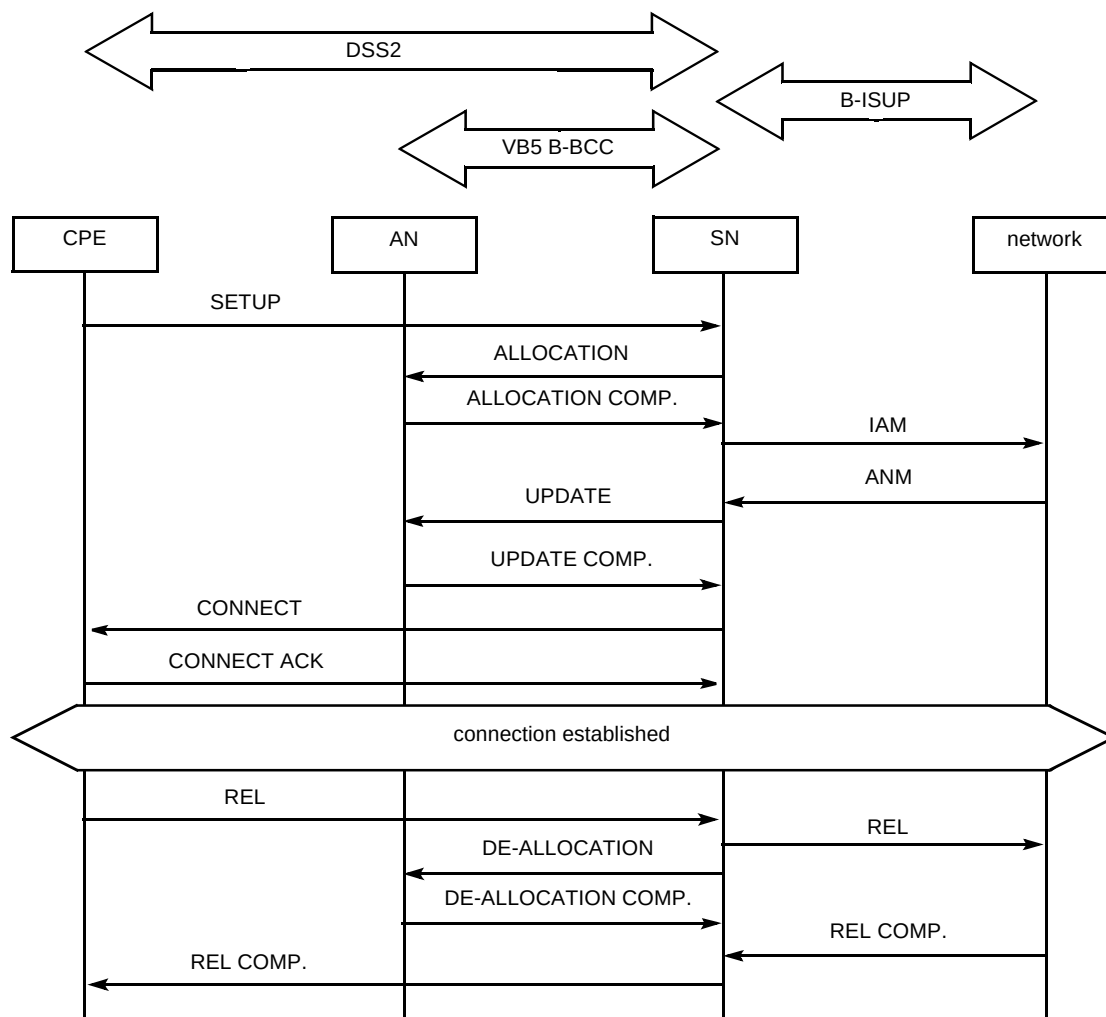


Fig 10 AN connection establishment/release — message sequence chart.

in the SN is more closely related to directly connected UNI accesses.

Figure 10, in its connection establishment and release message sequence chart, illustrates the use of allocation, update, and de-allocation messages and their relationship with DSS2 and B-ISUP.

5. VB5 management

The management (Q3) interfaces associated with access networks and service nodes which are linked with VB5 traffic interfaces need to support the configuration and fault and performance management functions associated with the VB5 interfaces. These Q3 interfaces are specified in terms of management models which are extensions of the ITU-T management model for ATM [15]. In a management implementation, managed objects which conform to the classes defined for the model, can be manipulated to perform the various management functions. The management models for the AN and the SN differ slightly; firstly the model for the AN is described (see Fig 11).

Transmission convergence (TC) functions both at UNIs and at VB5 interfaces are modelled by TC adaptor objects and connected VPs are modelled by contained VP connection termination point (CTP) objects. End-points of VP connections in the access network are represented by VP trail termination point (TTP) objects and connected VCs within these end-points are represented by contained VC

CTP objects. Any end-points of VC connections are represented by VC TTP objects.

Logical user port objects group VPs at user ports according to whether or not the VPs terminate in the AN. If they do not terminate in the access network, the intermediate points of VP connections at a user port are identified and labelled with VPCI values (pointer 2). If they do terminate, the end-points are identified and allocated VPCI values (pointer 3 of logical user port class).

When VP connections at the VB5 interface terminate in the AN, the logical service port objects group their own VP connection end-points and label these with VPCI values (pointer 3 of *IspVb51An* class). Logical service port objects also contain an RTMC communications path object (a communications path subclass) to identify the end-point of the VC connection for the RTMC protocol (pointer 5a) and its signalling adaptation profile (pointer 4a).

The modelling in the SN differs slightly from that in the AN because it takes account of user signalling (see Fig 12). If user VPs are only cross-connected in the SN then they are allocated VPCI labels by UNI access objects (pointer B) with a VP subclass used to allow the modelling of remote blocking. If they are terminated, it is the end-points which are labelled (pointer 8) and again a subclass is used to allow modelling of remote blocking. This VPCI labelling in the SN assumes that the VP connections are directly related to user accesses. If VP connections are not directly related to a user access (for instance, if there is VC cross-connection in

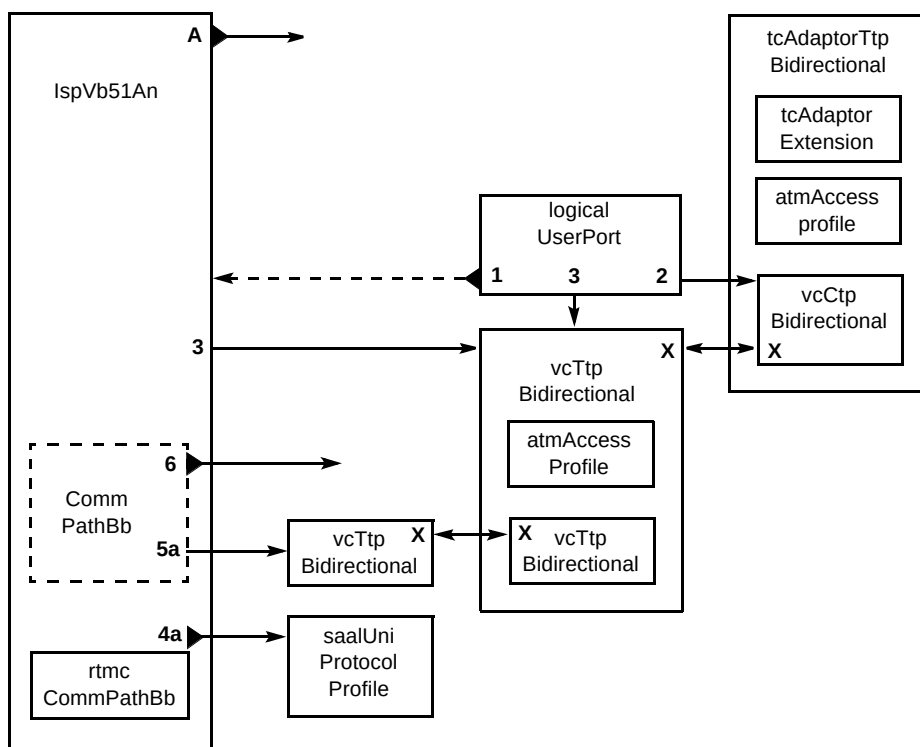


Fig 11 Object relationships in the access network.

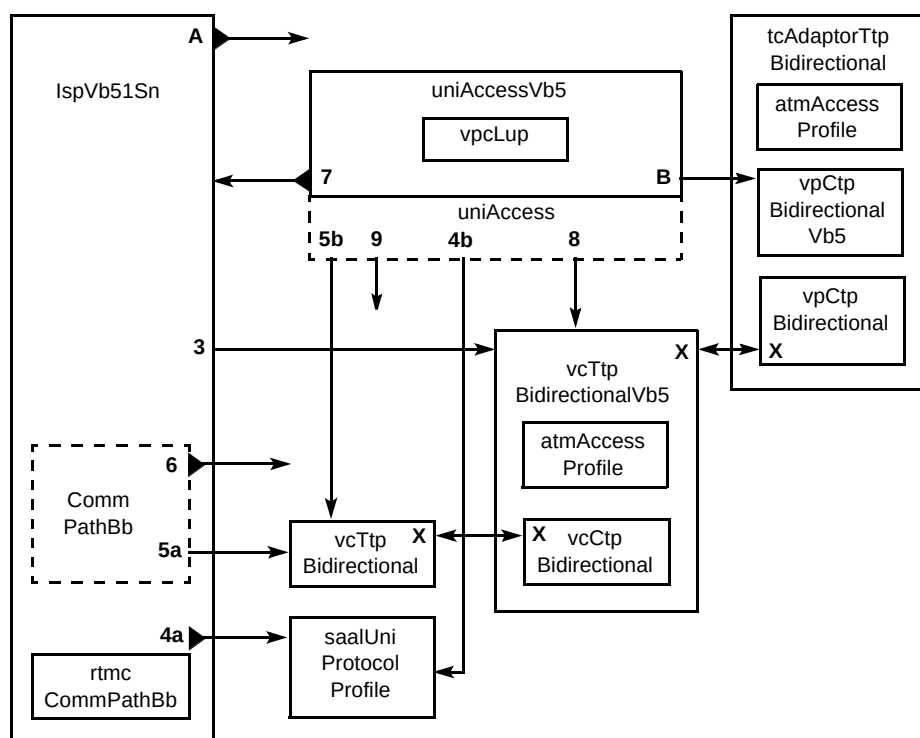


Fig 12 Object relationships in the service node.

the AN), their terminations are identified and labelled by LSP objects (pointer 3 of *lspVb51Sn* class).

The end-points of VC connections for user signalling are identified and profiles are assigned by UNI access objects in a similar way as for the RTMC protocol (pointers 5b and 4b). UNI access objects for indirect accesses also identify their logical service port (pointer 7) and contain VPC LUP objects to represent the remote blocking of any VP connections at the remote user port which are terminated in the AN.

The objects in the AN and SN models have data attributes (which the operations system can read and often write to), actions (which are requests sent by the operations system and which may trigger replies), and notifications (which can be sent to the operations system autonomously by the AN or SN). However, it is not appropriate to describe these here and the interested reader is referred to the VB5 management specifications [16, 17] for further details of these and how they are related to the information flows across VB5 interfaces.

6. Conclusions

The availability of a standardised SNI, which the VB5 interface provides, enables open procurement of broadband access networks. This should advance their development and deployment, with a consequent impact on delivery of broadband communications. This is particularly significant for the more cost-sensitive market sectors, such as residential and small business.

Progress in developing VB5 standards is now well advanced. In ETSI the VB5.1 interface specification finished its public enquiry phase in October 1997. In the ITU-T, work originally done by ETSI on VB5.1 has been incorporated into the ITU-T Recommendation G.VB51, which has been submitted for approval.

Work in ETSI and the ITU-T on the VB5.2 interface is running concurrently with good co-operation. A stable draft has been submitted for approval in April 1998.

Management specifications will follow shortly after their respective interface specifications.

Other standards bodies have included references to VB5 in their documents rather than attempting to derive their own standard for the SNI, in particular:

- the Digital Audio-Visual Council (DAVIC) A4 reference point [18] maps directly on to the ETSI/ITU SNI and uses VB5 protocols,
- the Full Services Access Network (FSAN) Group [19] has specified VB5 as the interface from the AN to the core network,
- the ATM Forum Residential Broadband Working Group has adopted VB5 as one option for their access network interface [20].

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